

helix-ui (Flutter app shell)

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Master technical specification — Volume 4 (Clients), nano-detail deep-dive. This document **deepens** the *Flutter helix-ui codebase* part of the pass-1 client overview [04_CLI §1/§2/§6/§8] into an implementation-ready specification of the **app shell**: the melos monorepo (its five packages + three apps), the runHelixApp(flavor, home, capabilities) wiring point, capability-gating / tree-shaking, the Dart-facing FFI wrapper, the TunnelPlatform MethodChannel + EventChannel contract, the Riverpod app-shell state machine, app lifecycle (foreground/background/permission), and the test points tied to the §11.4.169 closed test-type vocabulary. SPEC-ONLY: it describes **what to build**, not the shipping product.

Boundary with sibling docs. This document **consumes** (a) the *exported* Rust FFI surface owned by the pass-1 client doc [04_CLI §3] — start/stop/status_stream/exits/set_exit/set_shields/advertise/attach_tun; and (b) the **core-emitted TunnelStatus enum** owned by Volume 2's orchestrator [02_ORCH §4.1]. It **owns** everything Dart-side from the generated bindings upward: the package graph, the wiring, the flavor/capability model, the Dart-side TunnelPlatform impls' contract, the UI state projection, and app lifecycle. It does **not** own the Rust internals (Volume 1/2), the per-platform *native* shim bodies (Volume 4 shim doc — it owns only the Dart side of the channel), nor the control-plane wire (Volume 3).

Evidence base. Citations inline by id: [04_CLI §N] = final/03-client-core-and-ui.md (the pass-1 overview this deepens); [04_UI] = 04_VPN_CLD/HelixVPN-helix-ui-Flutter.md; [04_ARCH §N] = 04_VPN_CLD/HelixVPN-Architecture-Refined.md; [04_P0 §N] = ... Phase0-Spike.md; [02_ORCH §N] = final/v02-data-plane/orchestrator-and-state.md (canonical TunnelStatus); [02_TX §N] = final/v02-data-plane/transport-trait.md; [research-flutter_ffi] = v09-research/research-flutter_ffi.md; [research-ios_android] = the per-platform shim research; [SYN §N] = cross-document synthesis; [05_YB0] = operator mandate. Any claim not grounded in the evidence base is tagged UNVERIFIED per constitution §11.4.6 — never fabricated.

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0. Position, ownership, and invariants

0.1 What this document owns

flowchart TB

```
    subgraph Owns["OWNED HERE – the Flutter app shell (Dart)"]
      direction TB
      MELOS["melos workspace<br/>5 packages + 3 apps + tool/"]
      WIRE["runHelixApp(flavor, home, capabilities)<br/>+
      capability gating / tree-shaking"]
      FFIW["helix_core_ffi Dart wrapper<br/>(over
      frb_generated.dart)"]
      TPDART["TunnelPlatform Dart contract<br/>(MethodChannel +
      EventChannel)"]
      STATE["app-shell state machine<br/>(Riverpod projection of
      TunnelStatus)"]
      LIFE["app lifecycle + permission flow"]
    end
    subgraph Consumes["CONSUMED – frozen elsewhere"]
      direction TB
      FFISURF["Rust FFI surface [04_CLI §3]<br/>start/stop/
      status_stream/exits/..."]
      TS["core TunnelStatus enum [02_ORCH §4.1]<br/>5 variants –
      CANONICAL"]
      NATIVE["native shim bodies (Volume 4 shim doc)<br/>Swift/
      Kotlin/C#/C++/ArkTS"]
      API["helix_api REST/WS (Volume 3)"]
    end
    MELOS --> WIRE --> FFIW --> FFISURF
    FFIW --> TS
    TPDART -. "channel" .-> NATIVE
```

STATE --> FFIW
STATE --> API

This document owns six Dart-side contracts and nothing else:

#	Contract	Owned here	Consumed from
A1	melos workspace shape — the five packages, three apps, their pubspec dependency graph, the codegen/tool/scripts	§1–§2	—
A2	Dart FFI wrapper — the idiomatic HelixCore abstract class over <code>frb_generated.dart</code> ; the Dart <code>TunnelStatus</code> mirror (kept byte-consistent with [02_ORCH §4.1])	§3	Rust surface [04_CLI §3]; <code>TunnelStatus</code> [02_ORCH §4.1]
A3	TunnelPlatform channel contract (Dart side) — the <code>MethodChannel("helixvpn/tunnel")</code> + <code>EventChannel("helixvpn/tunnel/events")</code> method names, argument codecs, <code>TunnelConfig/PlatformTunnelEvent</code> serialization	§4	native shim bodies (Volume 4 shim doc)
A4	Flavor + capability model — <code>runHelixApp</code> , <code>HelixFlavor</code> , <code>Capability</code> , <code>NoCore</code> null impl, the tree-shake mechanism	§5–§6	—
A5	App-shell state machine — <code>UiConnectionState</code> projection of the core <code>TunnelStatus</code> ; the <code>ConnectController</code> <code>AsyncNotifier</code> ; the "UI is a pure function of the status stream" rule	§7	<code>helix_api</code> (Volume 3)
A6	App lifecycle + permission flow — <code>AppLifecycleState</code> handling, the permission state machine, web-no-core path	§8	OS lifecycle APIs

0.2 Non-negotiable invariants (inherited + tightened)

#	Invariant	Source	App-shell tightening
CI1	<code>helix_core_ffi</code> owns logic + status; it does NOT own the OS tunnel lifecycle — that is the <code>TunnelPlatform</code> shim.	[04_CLI CI1]	§3 wrapper exposes no "create TUN" verb; only <code>attachTun(fd)</code> (called by the shim). §4.
CI2	The UI is a pure function of the core's status stream. No polling.	[04_CLI CI2, 04_UI §4.2]	§7: the only writer of <code>UiConnectionState</code> is the <code>tunnelStatusProvider</code> fold; widgets watch, never read-and-poll.
CI3	Console builds to Web; Access/Connector do not. Console depends on	[04_CLI CI3,	§5–§6: capability gating + conditional imports make

#	Invariant	Source	App-shell tightening
	helix_api <i>only</i> — no helix_core_ffi, no dart:ffi.	04_ARCH §5.7]	dart:ffi unreachable on the Console/Web build.
CI5	State must be announced, not just colored.	[04_CLI CI5, 04_UI §9]	§7.4: every UiConnectionState carries a semanticLabel; the ConnectButton Semantics reads it.
AS-I7	One core TunnelStatus enum on the wire (5 variants). The Dart frb mirror MUST match [02_ORCH §4.1] byte-for-byte; any richer UI state (disconnected, danger, path) is a Dart-side projection computed in §7.2, never a fabricated extra wire variant.	derived from [02_ORCH §4.1] + §11.4.6	§3.3, §7.2.
AS-I8	Native $\Rightarrow$ Dart channel is the only untyped seam and is covered by a per-OS real-device smoke (never a mock-claiming-PASS).	[04_CLI O5, §11.4.27]	§4.1 O1–O6; §11 IT/SC/MEM.
AS-I9	Capability absent $\Rightarrow$ code tree-shaken out. A flavor that lacks Capability.tunnel MUST NOT link the Rust staticlib (so Console reaches Web).	[04_UI §2/ §8]	§5.3, §6 (the conditional-import mechanism, not a runtime if).

0.3 The reconciliation this document fixes (TunnelStatus)

The pass-1 overview [04_CLI §3.1] sketched a **7-variant** client mirror that added Disconnected, a path field on Connected, and Danger{kind}. Volume 2's orchestrator [02_ORCH §4.1] — the code that actually *emits* the stream — froze the enum at **5 variants** (Connecting | Handshaking | Connected{transport, rtt_ms} | Reconnecting | Down{reason}). Per AS-I7 and §11.4.6, this document resolves the divergence:

- **The frb mirror IS the 5-variant core enum** (§3.3) — byte-for-byte, asserted by the FFI contract test (§11 UT).
- **disconnected and danger are Dart-side derived UI states** (§7.2): a pure project(AsyncValue<TunnelStatus>) -> UiConnectionState function. disconnected is the synthetic pre-stream / post-Down{reason:"stopped"} state; danger is derived by **classifying Down.reason** against the closed vocabulary of [02_ORCH §4.4] (auth-failed/ladder-exhausted/host-fatal/no-route/pinned-transport-failed $\Rightarrow$ danger; stopped $\Rightarrow$ neutral disconnected).

- **path (direct vs relay) is a proposed Phase-2 core extension**, NOT present in the v1 mirror — surfaced as decision **D-UI-3** (§13), UNVERIFIED until the core adds it; the StatusChip renders path only when present.

This keeps the wire enum consistent with [02_ORCH §4.1] while still delivering the overview's UX (CI5, the red-on-danger palette).

1. The melos monorepo (helix-ui/)

helix-ui/ is a melos-managed multi-package Dart workspace [04_UI §1, 04_CLI §2]. Decoupling per §11.4.28/§11.4.74: helix_design, helix_core_fffi, helix_api, and helix_l10n are independently publishable / reusable; project-specific glue lives only in helix_domain and the three apps/.

1.1 Tree layout

```
helix-ui/
├─ melos.yaml                # workspace task runner
(bootstrap/version/run-everywhere)
├─ pubspec.yaml              # workspace root
(dev_dependencies: melos)
├─ packages/
│   └─ helix_design/         # tokens · theme · widgets ·
icons · motion → reusable
│   └─ helix_core_fffi/      # frb bindings to helix-core
(Access/Connector only)
│   │   └─ lib/
│   │   └─ frb_generated.dart # GENERATED – do not hand-
edit
│   │   └─ helix_core.dart   # idiomatic wrapper over the
generated API (§3)
│   │   └─ tunnel_platform.dart # TunnelPlatform channel
contract (§4)
│   │   └─ src/no_core.dart   # NoCore null impl
(Console/web; §5.2)
│   │   └─ rust/ -> ../../../../helix-core/crates/helix-ffi #
symlink to the Rust FFI crate
│   └─ helix_api/            # generated OpenAPI REST
client + WS/SSE client (all apps)
│   └─ helix_domain/         # models, Riverpod providers,
use-cases, runHelixApp() (§5)
│   └─ helix_l10n/           # ARB localization (en, ru,
zh-Hans, ...)
├─ apps/
│   └─ access/               lib/main_access.dart + ios/ macos/ android/
windows/ linux/ ohos/ aurora/
│   └─ connector/           lib/main_connector.dart + lib/
main_connector_headless.dart (no UI)
│   └─ console/             lib/main_console.dart (the ONLY flavor
that builds to Web)
```

```
└─ tool/                                # codegen scripts (frb,
openapi, intl) + drift checks
```

1.2 Package dependency graph (the decoupling contract)

flowchart BT

```

    design["helix_design<br/>(tokens · widgets)"]
    l10n["helix_l10n<br/>(ARB)"]
    api["helix_api<br/>(REST/WS – Volume 3)"]
    ffi["helix_core_ffi<br/>(frb + TunnelPlatform)"]
    domain["helix_domain<br/>(providers · runHelixApp)"]

    access["apps/access<br/>Capability.
{tunnel,account,splitTunnel}"]
    connector["apps/connector<br/>Capability.
{tunnel,advertise,localAcl}"]
    console["apps/console<br/>Capability.{admin} – NO tunnel"]

    domain --> design
    domain --> l10n
    domain --> api
    domain -. "conditional (CI3)" .-> ffi

    access --> domain
    access --> ffi
    connector --> domain
    connector --> ffi
    console --> domain
    console -. "NO ffi edge (web)" .-> ffi

    classDef no fill:#fee,stroke:#c33;
    class console no
```

The load-bearing edges: helix_domain depends on helix_core_ffi only through a **conditional import** (§6.1), so the Console app — which never adds the ffi dependency to its pubspec.yaml — pulls in the NoCore stub instead and never links dart:ffi / the Rust staticlib (CI3, AS-I9). Every other package edge is unconditional.

1.3 melos.yaml (workspace scripts)

```

# helix-ui/melos.yaml
name: helix_ui
packages:
  - packages/**
  - apps/**
command:
  bootstrap:
    runPubGetInParallel: true
scripts:
  analyze: { exec: "dart analyze --fatal-infos", concurrency:
6 }
```

```

test:      { exec: "flutter test", packageFilters: { dirExists:
             test } }
gen:      # codegen – run on any Rust-API / OpenAPI /
           ARB change
run: |
  dart run tool/frb_generate.dart          #
      flutter_rust_bridge_codegen generate
  dart run tool/openapi_generate.dart      # OpenAPI ->
      helix_api/lib/rest_generated.dart
  flutter gen-l10n --arb-dir packages/helix_l10n/lib/arb
gen-check: # CI/pre-tag drift gate (§11 UT):
           regenerate, fail if `git diff` non-empty
run: "melos run gen && git diff --exit-code"

```

`melos run gen-check` is the **no-drift guarantee** [04_UI §11, 04_ARCH §4.2]: a Rust FFI-surface change that is not regenerated, or an OpenAPI change the Dart client lags, is a build-blocking finding — the apps can never silently diverge from the core or the control plane (§11.4.156: this runs in the local pre-tag sweep, not a hosted runner).

1.4 Versioning & per-package pins (research-pinned)

Package	Key dep	Pinned version	Source
helix_core_ffi	flutter_rust_bridge	2.12.0 (Dart dep == bridge-crate version)	[research-flutter_ffi §1/§7]
helix_domain	flutter_riverpod	3.3.2 (Riverpod 3.x line)	[research-flutter_ffi §6]
helix_design	Flutter SDK	mainline (Material 3)	[04_UI §3]
apps/access (ohos)	OpenHarmony SIG fork	pin per fork (lags mainline)	[research-flutter_ffi §5]
apps/access (aurora)	OMP fork flutter-aurora	pin per fork	[research-flutter_ffi §4]

Pin discipline (§11.4.6). The frb Dart-dep version **MUST** equal the bridge-crate version or codegen output mismatches at runtime [research-flutter_ffi §1]. The two forked targets pin a *specific* fork commit (HarmonyOS/Aurora lag mainline Flutter) and build on **isolated local runners** so a fork lag never blocks a mainline release (§12 Phase 3, [04_UI §6.1]).

2. The five packages — public surfaces

2.1 helix_design (tokens · theme · widgets)

Reusable, project-agnostic. Owns the connection-state palette mapped to the core TunnelStatus and the signature components [04_CLI §7].

```
// helix_design/lib/tokens.dart
class HelixTokens {
  static const stateDisconnected = Color(0xFF6B7280); // neutral
  static const stateConnecting   = Color(0xFFF59E0B); // amber,
  static const stateConnected    = Color(0xFF10B981); // green,
  static const stateDanger       = Color(0xFFEF4444); // red
  static const brandSeed         = Color(0xFF3B5BDB); // drives M3 ColorScheme, per-role
  // spacing 4-base {4,8,12,16,24,32,48}; radius
  // motion {fast120,base220,slow360}; elevation 0..3 (flat,
  // color-as-hierarchy).
}

// helix_design/lib/theme.dart – the state→color map the whole app
// reads (consumes §7.2 UiConnectionState)
extension HelixThemeX on BuildContext {
  Color colorForUiState(UiConnectionState s) => switch (s) {
    UiDisconnected() => HelixTokens.stateDisconnected,
    UiConnecting()   => HelixTokens.stateConnecting,
    UiConnected()    => HelixTokens.stateConnected,
    UiReconnecting() => HelixTokens.stateConnecting,
    UiDanger()       => HelixTokens.stateDanger,
  };
}
```

Public widget surface (consumed by helix_domain/apps): ConnectButton, StatusChip, ExitPicker, ShieldIndicator, NetworkTile, PolicyEditor, AdaptiveScaffold [04_CLI §7.2]. AdaptiveScaffold branches on **width**, never Platform.isX (CI3 corollary — a resized desktop window behaves like a phone; web "just works") [04_UI §7]. helix_design depends on no other Helix package (pure design system).

2.2 helix_core_ffi (frb + TunnelPlatform)

Owns the generated bindings + the idiomatic wrapper + the TunnelPlatform Dart contract + the NoCore null impl. Detailed in §3–§4. Depends only on flutter + flutter_rust_bridge + helix_l10n (for error strings). **The only package that references dart:ffi** — gated behind a conditional import so it is unreachable on web (§6.1).

2.3 helix_api (REST/WS — Volume 3 client)

Generated OpenAPI Dart client (rest_generated.dart) + a hand-written WS/SSE client (ws_client.dart) for the control-plane event stream (device.online, route.changed, policy.compiled, device.revoked) [04_CLI §8.3]. Same generated model types are used by the test fakes (§11 UT). Depends on no Helix package (pure transport client). **This is the only dependency the Console/Web build needs** (CI3).

2.4 helix_domain (providers · use-cases · runHelixApp)

The project-specific glue: Riverpod providers (tunnelStatusProvider, connectControllerProvider, flavorProvider, capabilitiesProvider, helixCoreProvider, tunnelPlatformProvider), use-cases, models, and the runHelixApp wiring (§5). Depends on helix_design, helix_l10n, helix_api, and — **conditionally** — helix_core_ffi.

2.5 helix_l10n (ARB localization)

Flutter intl/ARB. First-tier locales: **en** (baseline), **ru** (Aurora market), **zh-Hans** (HarmonyOS market); RTL-ready [04_UI §9]. Censorship-region users are core users, so clear localized connection state matters most here. Generated via flutter gen-l10n (wired into melos run gen, §1.3). Depends on nothing.

3. The Dart FFI surface (helix_core_ffi)

3.1 The generation pipeline (research-flutter_ffi)

flowchart LR

```
rs["helix-ffi/src/api.rs<br/>(hand-authored Rust surface,
[04_CLI §3.1)]"]
frb["flutter_rust_bridge_codegen<br/>generate (v2.12.0)"]
gen["frb_generated.dart<br/>+ frb_generated.rs (Rust glue)"]
wrap["helix_core.dart<br/>(idiomatic HelixCore wrapper)"]
rs --> frb --> gen --> wrap
frb -. "also (UniFFI, native shims)" .-> uni["Swift/Kotlin
bindings<br/>(Volume 4 shim doc, D-CLIENT-3)"]
```

flutter_rust_bridge v2.12.0 [research-flutter_ffi §1/§7] is a *codegen* binding generator: it reads the annotated Rust helix-ffi crate and emits the Dart FFI glue + a Rust wrapper. v2 supports `async fn`, opaque `RustOpaque` handles, and — load-bearing for this VPN core — **StreamSink<T> → Dart Stream<T>** for the status channel (frb issue #347, solved) [research-flutter_ffi §1]. UniFFI is used *additionally* for the native shims where a platform extension links the core directly (Swift/Kotlin) — `uniffi-dart` itself is not production-ready, so frb owns the Dart side and UniFFI owns only the native-shim side (decision D-CLIENT-3) [research-flutter_ffi §2].

3.2 The Dart-facing wrapper (helix_core.dart)

frb mirrors every `#[frb(mirror)]` Rust type into an identical Dart type — the Dart TunnelStatus/Shields/ExitOption are the *generated* mirrors, never hand-written parallels. The wrapper adds ergonomics only [04_CLI §3.2]:

```
// helix_core_ffi/lib/helix_core.dart – idiomatic facade over
// frb_generated.dart
abstract interface class HelixCore {
  /// Logic-only start (NOT the OS tunnel; the shim owns that –
  /// CI1).
```

```

Future<void> start({required String transport, String?
    mapPathOrSession,
    CoreMode mode = CoreMode.client});
Future<void> stop();

/// frb StreamSink<TunnelStatus> → broadcast Dart Stream (NO
polling – CI2).
Stream<TunnelStatus> statusStream();

Future<List<ExitOption>> exits();
Future<void> setExit(String id, {List<String>? multiHopChain});
Future<void> setShields(Shields s);

// connector mode:
Future<AdvertiseResult> advertise(List<String> cidrs);

// shim handoff – called BY the TunnelPlatform shim (§4), NOT by
UI widgets:
Future<void> attachTun(int fd);
Future<void> detachTun();
}

/// The real, frb-backed impl (Access/Connector). Wraps
frb_generated.dart calls.
final class RealHelixCore implements HelixCore { /* delegates to
    frb generated API */ }

```

The happy path a client app drives (mirrors the Phase-0 G5 demo [04_P0 §9]):

```

await core.start(transport: 'auto', mapPathOrSession:
    session.token);
core.statusStream().listen(uiFold); // Riverpod fold, §7.1
// ...later:
await core.stop();

```

3.3 The TunnelStatus Dart mirror (canonical — AS-I7)

The frb mirror MUST equal [02_ORCH §4.1] byte-for-byte. frb renders a Rust enum with data variants as a Dart **sealed class** hierarchy:

```

/// helix_core_ffi/lib/frb_generated.dart (GENERATED – shown for
the contract; do not hand-edit)
sealed class TunnelStatus {
    const TunnelStatus();
}
final class Connecting extends TunnelStatus { const
    Connecting(); }
final class Handshaking extends TunnelStatus { const
    Handshaking(); }
final class Connected extends TunnelStatus {
    final String
        transport; // Transport::kind() e.g. "plain-udp" |
        "masque-h3" [02_TX §2]

```

```

final int    rttMs;      // WG-handshake RTT EWMA (u32 → Dart
                        int)
const Connected({required this.transport, required this.rttMs});
}
final class Reconnecting extends TunnelStatus { const
    Reconnecting(); }
final class Down extends TunnelStatus {
    final String reason;    // stable-prefix vocabulary [02_ORCH
                        §4.4]
    const Down({required this.reason});
}

```

Consistency assertion (§11 UT — the FFI contract test). A shared fixture corpus of (Rust TunnelStatus value, expected Dart value) pairs proves the mirror matches the Rust source byte-for-byte; a drift FAILs the build (§1.3 gen-check). The mirror has **exactly five** variants — no Disconnected, no Danger, no path on the wire (AS-I7). Those are derived in §7.2.

3.4 The remaining mirrored types

```

// GENERATED mirrors of [04_CLI §3.1] ([frb(mirror)] Rust types):
enum CoreMode { client, connector }

final class Shields {
    final bool killSwitch, dnsProtection, daita, postQuantum;
    final List<String> splitTunnel; // per-route bypass; per-app
    // handled in the shim layer
    const Shields({required this.killSwitch, required
        this.dnsProtection,
        required this.daita, required this.postQuantum,
        this.splitTunnel = const []});
}
final class ExitOption {
    final String id, kind, label;    // kind = "privacy_exit" |
    // "network"
    final String? country, jurisdiction;
    final int? rttMs;
    const ExitOption({required this.id, required this.kind,
        required this.label,
        this.country, this.jurisdiction, this.rttMs});
}
final class AdvertiseResult { final List<String> accepted,
    conflicts;
    const AdvertiseResult({required this.accepted, required
        this.conflicts}); }

```

3.5 Stream resource discipline (Riverpod 3.x)

The frb `Stream<TunnelStatus>` is wrapped in a `StreamProvider.autoDispose` (§7.1) so the underlying Rust `StreamSink` subscription is torn down when no UI listens [research-flutter_ffi §6]. **Riverpod 3.0 caveat:** `StreamProvider` *pauses* its `StreamSubscription` when not actively listened — relevant because a paused

subscription means the Rust side may stop being polled. For the *foreground connected* screen this is correct (always listened); for a backgrounded app the pause is desirable (§8.3). The provider is `autoDispose` so a non-`autoDispose` "almost never destroyed" leak is avoided [research-flutter_ffi §6].

4. The TunnelPlatform channel contract (Dart side)

Every native shim implements **one contract** and does **only three things**: configure the OS tunnel, hand packets to/from `helix-core`, and report lifecycle [04_CLI §4]. This document owns the **Dart side** of that channel (the abstract class + the codec); the native bodies (Swift/Kotlin/C#/C++/ArkTS) are Volume 4's shim doc.

```
// helix_core_ffi/lib/tunnel_platform.dart
// One MethodChannel ("helixvpn/tunnel") + one EventChannel
// ("helixvpn/tunnel/events").
abstract interface class TunnelPlatform {
  /// OS asks permission, creates the TUN, links the core, starts
  /// the pump.
  Future<void> startTunnel(TunnelConfig cfg);
  /// Tear down: detach core, destroy TUN, release permission hold
  /// (04).
  Future<void> stopTunnel();
  /// Lifecycle ONLY – NOT data. up/down/permissionDenied/revoked/
  /// error.
  Stream<PlatformTunnelEvent> events();
}

final class TunnelConfig {
  final String overlayIp;           // e.g. 100.64.0.7/32 (or
  // fd7a::.../128 v6 overlay)
  final List<String>
    routes;           // AllowedIPs the OS should send into
  // the tunnel
  final List<String> dnsServers;    // tunnel DNS (kill
  // plaintext :53 off-tunnel)
  final List<String> splitExcludeApps; // Android/desktop per-app
  // bypass (platform-handled)
  final int mtu;           // negotiated; 1280 over
  // MASQUE, 1420 plain WG [02_TX §4.8]
  final String sessionOrMapToken;   // handed straight to
  // helix_core_start
  const TunnelConfig({required this.overlayIp, required
    // this.routes,
    required this.dnsServers, this.splitExcludeApps = const [],
    // this.mtu = 1280,
    required this.sessionOrMapToken});
}

enum PlatformTunnelEventKind { up, down, permissionDenied,
  // revoked, error }
final class PlatformTunnelEvent {
  final PlatformTunnelEventKind kind;
```

```

final String? detail; // honest reason on error/
                      revoked (§11.4.6: no guessing)
const PlatformTunnelEvent(this.kind, [this.detail]);
}

```

4.1 Contract obligations (every shim MUST satisfy)

#	Obligation	Why
O1	startTunnel is idempotent + permission-aware : an OS VPN-permission denial emits permissionDenied (not error) and leaves no half-open TUN (§11.4.14).	clean UX, no leaks
O2	The shim hands the core a packet fd / pump ; it never crypto/obfuscates. helix-core is the only place WG + transport live [02_TX I4].	one core, no fork
O3	events() emits revoked when the OS (or admin via device.revoked, Volume 3) kills the tunnel out-of-band — the UI must reflect it, never claim "connected".	CI2 honesty
O4	stopTunnel restores the OS to a quiescent state (no orphan routes, no leaked DNS, kill-switch rules removed) on every exit path.	§11.4.14
O5	The shim is the only untyped seam (native $\rightleftharpoons$ Dart) — covered by a per-OS real-device smoke, never a mock-claiming-PASS (AS-I8).	§11.4.27
O6	The EventChannel stream is at-least-once + idempotent : a duplicate up/down is collapsed by the UI fold (§7.3); a torn re-listen after backgrounding re-emits the latest event.	§8.3 robustness

4.2 Channel wire (method names + argument codec)

The MethodChannel uses the **StandardMethodCodec** (default), so TunnelConfig is marshalled as a Map<String, Object?>:

Direction	Channel	Method / event	Payload
Dart → native	helixvpn/tunnel	startTunnel	{overlayIp, routes:[...], dnsServers:[...], splitExcludeApps:[...], mtu, sessionOrMapToken}
Dart → native	helixvpn/tunnel	stopTunnel	null
native → Dart	helixvpn/tunnel/events	(stream)	{kind:"up" "down" "permissionDenied" "revoked" detail?}

```

// helix_core_ffi/lib/src/channel_tunnel_platform.dart – the
// default MethodChannel impl
final class ChannelTunnelPlatform implements TunnelPlatform {
  static const _m = MethodChannel('helixvpn/tunnel');
  static const _e = EventChannel('helixvpn/tunnel/events');

  @override

```

```

Future<void> startTunnel(TunnelConfig c) =>
    _m.invokeMethod('startTunnel', {
        'overlayIp': c.overlayIp, 'routes': c.routes,
        'dnsServers': c.dnsServers,
        'splitExcludeApps': c.splitExcludeApps, 'mtu': c.mtu,
        'sessionOrMapToken': c.sessionOrMapToken,
    });

@override
Future<void> stopTunnel() => _m.invokeMethod('stopTunnel');

@override
Stream<PlatformTunnelEvent> events() => _e
    .receiveBroadcastStream()
    .map((m) => PlatformTunnelEvent(
        PlatformTunnelEventKind.values.byName((m as Map)
            ['kind'] as String),
        m['detail'] as String?,
    ));
}

```

4.3 The FFI $\rightleftharpoons$ shim seam (the bug-preventing division of labor)

```

sequenceDiagram
    participant UI as Flutter UI (Dart)
    participant DOM as helix_domain (Riverpod)
    participant FFI as helix_core_ffi (frb)
    participant SHIM as TunnelPlatform shim (native)
    participant OS as OS tunnel (NE / VpnService / wg-nt / tun)
    participant CORE as helix-core (Rust)

    UI->>DOM: connectController.toggle()
    DOM->>SHIM: startTunnel(TunnelConfig) %% MethodChannel
    SHIM->>OS: request permission + create TUN (overlay IP,
routes, DNS)
    OS-->>SHIM: tun fd / packetFlow
    SHIM->>CORE: helix_core_start(cfg) + attach_tun(fd) %%
UniFFI/JNI/C ABI
    SHIM-->>DOM: PlatformTunnelEvent.up %% EventChannel
    Note over CORE: pumps tun ↔ WG ↔ transport (Volume 2)
    CORE-->>FFI: TunnelStatus events (StreamSink)
    FFI-->>DOM: statusStream() → §7.2 project() → ConnectButton/
StatusChip
    UI->>FFI: core.setShields(...) / setExit(...) %% logic-only,
no OS tunnel (CI1)
    UI->>SHIM: stopTunnel()
    SHIM->>CORE: detach_tun() + helix_core_stop()
    OS-->>SHIM: tunnel torn down
    SHIM-->>DOM: PlatformTunnelEvent.down

```

The seam: **lifecycle commands flow UI → shim → core** (the OS owns the tunnel process), while **status events flow core → FFI → UI** (the core owns truth about protection state). setShields/setExit/exits are pure logic and go straight UI →

FFI without touching the OS tunnel (CI1). This is precisely the boundary that prevents "UI says connected while the OS tunnel is down": the UI believes only the core's statusStream, never its own intent (CI2; §7.5).

4.4 Native-shim signatures (the verbs the channel marshals to)

The Dart channel marshals to these per-platform entry points (bodies = Volume 4 shim doc; signatures fixed here so the Dart codec is implementable) [04_CLI §5, research-ios_android]:

```
// iOS/macOS – apps/access/ios/HelixTunnel/
// PacketTunnelProvider.swift (Swift)
final class PacketTunnelProvider: NEPacketTunnelProvider {
    override func startTunnel(options: [String:NSObject]?,
        completionHandler: @escaping (Error?) -> Void)
    override func stopTunnel(with reason: NEProviderStopReason,
        completionHandler: @escaping () -> Void)
}
// core entry (UniFFI/cbindgen): helix_core_start(_ token: String,
//     _ mode: CoreMode)
//                               helix_core_attach_tun(_ fd:
//                               Int32); helix_core_stop()

// Android – apps/access/android/.../HelixVpnService.kt (Kotlin)
class HelixVpnService : VpnService() {
    private external fun coreStart(sessionToken: String, fd:
        Int) // JNI → helix-core .so
    private external fun coreStop()
    override fun onStartCommand(intent: Intent?, flags: Int,
        startId: Int): Int // establish() → fd → coreStart
    override fun onRevoke() // 03: OS killed us → emit revoked
    override fun onDestroy() // 04: coreStop + stopForeground
}

// Aurora – apps/access/aurora/helix_tunnel_backend.cpp (C++; Qt/
// C++ backend)
extern "C" {
    int helix_core_start(const char* token, int mode); // C ABI
    // into the Rust cdylib
    int helix_core_attach_tun(int fd);
    void helix_core_stop(void);
}
class HelixTunnelBackend { /* opens tun, calls
    helix_core_attach_tun(fd), bridges via Friflex plugin
    */ };

// HarmonyOS NEXT – apps/access/ohos/.../HelixVpnAbility.ets
// (ArkTS → NAPI)
// ArkTS VpnExtensionAbility; NAPI bridges to the Rust .so;
// MethodChannel bridges ArkTS↔Flutter.
import vpnExtension from '@ohos.net.vpnExtension';
// native (N-API): helixCoreStart(token: string, mode: number):
// void; helixCoreAttachTun(fd: number): void;
```



```
// Windows – HelixTunnelSvc (C#, SYSTEM service) + Rust core in-
// service.
// Flutter side is a TunnelPlatform impl marshalling startTunnel/
// stopTunnel/events over a NAMED PIPE
// (\\.\pipe\helixvpn) – the Dart channel impl is
// WindowsPipeTunnelPlatform, not the default MethodChannel.
```

Windows / Web are the two non-default Dart impls. Windows uses a named-pipe WindowsPipeTunnelPlatform (the Flutter app is unprivileged; the SYSTEM service hosts the core) [04_CLI §5.3]. Web uses NoTunnelPlatform (throws if invoked — Console never tunnels, CI3, §6.2). Every other platform uses the default ChannelTunnelPlatform (§4.2). The impl is selected in runHelixApp (§5.2).

5. runHelixApp + capability gating + tree-shaking

5.1 The three entrypoints (one tree, three flavors)

```
// apps/access/lib/main_access.dart
void main() => runHelixApp(
  flavor: HelixFlavor.access,
  home: const ConnectScreen(),
  capabilities: const {Capability.tunnel, Capability.account,
    Capability.splitTunnel},
);

// apps/connector/lib/main_connector.dart
void main() => runHelixApp(
  flavor: HelixFlavor.connector,
  home: const ConnectorDashboard(),
  capabilities: const {Capability.tunnel, Capability.advertise,
    Capability.localAcl},
);

// apps/console/lib/main_console.dart – the ONLY flavor that
// builds to Web
void main() => runHelixApp(
  flavor: HelixFlavor.console,
  home: const ConsoleShell(),
  capabilities: const {Capability.admin}, // NO
    Capability.tunnel → no core_ffi linked (CI3)
);
```

5.2 The wiring point (helix_domain/lib/run_helix_app.dart)

```
enum HelixFlavor { access, connector, console }
enum Capability { tunnel, account, splitTunnel, advertise,
  localAcl, admin }

Future<void> runHelixApp({
  required HelixFlavor flavor,
```



```

required Widget home,
required Set<Capability> capabilities,
}) async {
  WidgetsFlutterBinding.ensureInitialized();
  final container = ProviderContainer(overrides: [
    flavorProvider.overrideWithValue(flavor),
    capabilitiesProvider.overrideWithValue(capabilities),
    // Capability.tunnel ⇒ a real frb-backed core; else the NoCore
    // stub (Console/web):
    helixCoreProvider.overrideWith((ref) =>
      capabilities.contains(Capability.tunnel) ?
        makeRealHelixCore() : const NoCore()),
    // platform impl selected once, here:
    tunnelPlatformProvider.overrideWith((ref) =>
      capabilities.contains(Capability.tunnel) ?
        makeTunnelPlatform() : const NoTunnelPlatform()),
  ]);
  runApp(UncontrolledProviderScope(
    container: container,
    child: HelixApp(home: home, theme: helixTheme(flavor)), //
    // theme per role (§2.1)
  ));
}

```

makeRealHelixCore() and makeTunnelPlatform() are resolved through the **conditional import** of §6.1, so on the Console/Web build they resolve to the NoCore/NoTunnelPlatform stubs and dart:ffi / the Rust staticlib is never referenced (CI3, AS-I9).

```

// helix_core_fffi/lib/src/no_core.dart – the null impl (Console/
// web)
final class NoCore implements HelixCore {
  const NoCore();
  @override Stream<TunnelStatus> statusStream() => const
    Stream.empty();
  @override Future<void> start({...}) async => throw const
    UnsupportedTunnelError();
  // every tunnel verb throws UnsupportedTunnelError; exits()/
  // advertise() likewise.
}

```

5.3 Capability → compiled-in map (the tree-shake table)

Capability	Compiles in	In flavor
tunnel	helix_core_fffi, TunnelPlatform shim, ConnectButton, ShieldIndicator, StatusChip	Access, Connector
account	account screen, OIDC + anonymous device-token enrollment, QR enroll	Access
splitTunnel	per-app/per-route bypass UI	Access
advertise	CIDR advertise UI, overlapping-CIDR conflict surface	Connector

Capability	Compiles in	In flavor
localAcl	site-local restriction editor	Connector
admin	tenants/devices/networks/policy/audit/topology + helix_api heavy tables	Console

flowchart TB

```

    cap{"Capability.tunnel<br/>present?"}
    cap -->|yes - Access/Connector| real["helixCoreProvider =
RealHelixCore<br/>tunnelPlatformProvider =
ChannelTunnelPlatform<br/>links Rust staticlib + dart:ffi"]
    cap -->|no - Console| nocore["helixCoreProvider = NoCore<br/>
>tunnelPlatformProvider = NoTunnelPlatform<br/>NO dart:ffi
reference → web-buildable"]
    real --> tree["Dart tree-shaker drops unused<br/>capability
widgets per flavor"]
    nocore --> web["console builds to Web (CanvasKit/Wasm)<br/>+
desktop/mobile admin"]

```

Why it tree-shakes (mechanism, not a runtime if). The Console app's `pubspec.yaml` does **not** declare a dependency on `helix_core_ffi`; the conditional import (§6.1) resolves the `makeRealHelixCore/` `makeTunnelPlatform` symbols to the stub library at compile time. Dart's AOT tree-shaker then has no reachable reference to `dart:ffi` or the `staticlib` from the Console entrypoint, so neither is linked [04_UI §2/§8]. Capability gating of *widgets* (e.g. `SplitTunnelScreen` only under `Capability.splitTunnel`) is a const-set membership the tree-shaker resolves when the screen is never reachable from a flavor's route table.

5.4 The connector headless entrypoint

Connector additionally ships `apps/connector/lib/main_connector_headless.dart` — the **same Rust core, no Flutter UI** — for appliance deployments [04_CLI §6, 04_ARCH §5.4]. It calls `helix_core_start(mode: connector)` directly (no `runApp`), is the *primary* Connector surface on appliances, and the Flutter Connector UI is the *optional* config surface (decision D-CLIENT-4). It links `helix_core_ffi` but not `helix_design`.

6. Which flavor builds to Web (Console only, no core_ffi)

6.1 The conditional-import mechanism

```

// helix_domain/lib/src/core_factory.dart
export
    'core_factory_stub.dart'          // default: throws /
    NoCore (used by web + Console)
    if (dart.library.ffi) 'core_factory_ffi.dart'; // dart:ffi
    present (native) → real core

```

```
// core_factory_ffi.dart (compiled only where dart:ffi exists –
// native targets)
HelixCore      makeRealHelixCore() => RealHelixCore();           //
// links frb + staticlib
TunnelPlatform makeTunnelPlatform() =>
    ChannelTunnelPlatform(); // (or WindowsPipe...)

// core_factory_stub.dart (compiled on Web AND when no tunnel
// capability)
HelixCore      makeRealHelixCore() => const NoCore();
TunnelPlatform makeTunnelPlatform() => const NoTunnelPlatform();
```

if (dart.library.ffi) is the Dart conditional-import guard: on the **Web** target `dart:ffi` is absent, so the stub is compiled and the Rust `staticlib` is never referenced. This is the *mechanical* reason the Console flavor — and only the Console flavor — builds to Web (CI3, AS-I9). Access/Connector build to Web would compile, but their `Capability.tunnel` core would be a `NoCore` stub — so they are **not shipped to Web** (a tunnel-less Access app is meaningless); only the Console targets Web by design.

6.2 The web Console surface (what Web actually is)

flowchart LR

```
web["console (Web build)"] --> api["helix_api ONLY<br/>(REST +
WS/SSE – Volume 3)"]
web -. -> |"N0"| ffi["helix_core_ffi / dart:ffi / Rust
staticlib"]
web --> proxy["optional in-page WASM MASQUE proxy<br/>
>(browser-scoped, Phase 3 – NOT a device VPN)"]
```

Browsers cannot open a TUN device [04_ARCH §5.7, CI3]. The Web build = **Helix Console (management) + account/config**, *plus* an optional in-page WASM MASQUE client that proxies *the browser's own* traffic to a joined network — **not** a system-wide tunnel (Phase 3, decision D-CLIENT-5). State this plainly to users: "fully responsive web app" = the Console and a browser-scoped proxy, never a device VPN. The Web flavor has **no TunnelPlatform implementation and no helix_core_ffi dependency**.

7. The app-shell state machine (Riverpod)

7.1 The providers (compile-safe DI)

Riverpod 3.x is the choice [04_CLI §8, 04_UI §4]: compile-safe DI, first-class turning of the FFI/WS streams into reactive UI, trivial test overrides, no BuildContext coupling (Bloc is the viable alternative — decision D-CLIENT-2).

```
// helix_domain/lib/providers/tunnel.dart
final flavorProvider = Provider<HelixFlavor>((_) => throw
    UnimplementedError()); // overridden in runHelixApp
final capabilitiesProvider = Provider<Set<Capability>>((_) =>
    throw UnimplementedError());
```

```

final helixCoreProvider = Provider<HelixCore>((_) => throw
    UnimplementedError());
final tunnelPlatformProvider = Provider<TunnelPlatform>((_) =>
    throw UnimplementedError());

// The raw core stream → reactive Dart state (NO polling – CI2):
final tunnelStatusProvider =
    StreamProvider.autoDispose<TunnelStatus>((ref) {
        return ref.watch(helixCoreProvider).statusStream(); // frb
        StreamSink → Dart Stream
    });

// The platform lifecycle event stream (up/down/permissionDenied/
// revoked/error):
final platformEventProvider =
    StreamProvider.autoDispose<PlatformTunnelEvent>((ref) {
        return ref.watch(tunnelPlatformProvider).events();
    });

// The single derived UI state the whole app renders (§7.2):
final uiConnectionStateProvider =
    Provider.autoDispose<UiConnectionState>((ref) {
        final status = ref.watch(tunnelStatusProvider);
        final pevent = ref.watch(platformEventProvider).valueOrNull;
        return projectUiState(status, pevent); // pure
        function, §7.2
    });

```

7.2 The UI-state projection (projectUiState — AS-I7)

The only place disconnected/danger are synthesized — a **pure function** of the 5-variant core enum + the latest platform event, never a fabricated wire variant:

```

// helix_domain/lib/src/ui_state.dart
sealed class UiConnectionState {
    String get semanticLabel; // CI5 – announced, never color-only
}
final class UiDisconnected extends UiConnectionState { String get
    semanticLabel => 'Not protected'; }
final class UiConnecting extends UiConnectionState { String get
    semanticLabel => 'Connecting'; }
final class UiConnected extends UiConnectionState {
    final String transport; final int rttMs; final String?
    path; // path: D–UI–3, null until core emits
    UiConnected(this.transport, this.rttMs, [this.path]);
    String get semanticLabel => 'Protected via $transport, $rttMs
    ms';
}
final class UiReconnecting extends UiConnectionState { String get
    semanticLabel => 'Reconnecting'; }
final class UiDanger extends UiConnectionState {
    final String kind; // "leak" | "killswitch_tripped" |
    "<Down.reason>"
    UiDanger(this.kind);
}

```

```

    String get semanticLabel => 'Not protected – $kind';
}

/// Maps the canonical 5-variant TunnelStatus (+ platform events)
to the UI palette.
UiConnectionState projectUiState(AsyncValue<TunnelStatus> s,
    PlatformTunnelEvent? pe) {
    // platform-side danger overrides core state immediately
    (revoke / OS kill):
    if (pe?.kind == PlatformTunnelEventKind.revoked)           return
        UiDanger('revoked');
    if (pe?.kind == PlatformTunnelEventKind.permissionDenied)
        return UiDisconnected();
    return switch (s.valueOrNull) {
        null            => UiDisconnected(),                    // pre-
        stream / no value yet
        Connecting()    => UiConnecting(),
        Handshaking()   => UiConnecting(),
        Connected(:final transport, :final rttMs) =>
            UiConnected(transport, rttMs),
        Reconnecting()  => UiReconnecting(),
        Down(:final reason) => _classifyDown(reason),           //
        §7.2.1
    };
}

UiConnectionState _classifyDown(String reason) => switch (reason)
{
    'stopped'            => UiDisconnected(),                    //
    clean, user-initiated (neutral)
    'auth-failed'        => UiDanger('auth-failed'),            //
    revoked/bad key – closed (§02_ORCH §8.5)
    'ladder-exhausted'   => UiDanger('ladder-exhausted'),
    'host-fatal'         => UiDanger('host-fatal'),
    'no-route'           => UiDanger('no-route'),
    'pinned-transport-failed' => UiDanger('pinned-transport-
        failed'),
    –                    => UiDanger(reason),                    //
    unknown reason → fail-safe to danger
};

```

7.2.1 The Down.reason classification source

The closed prefix vocabulary is owned by [02_ORCH §4.4]: stopped (clean) ⇒ UiDisconnected; everything else (auth-failed/ladder-exhausted/host-fatal/ no-route/pinned-transport-failed) ⇒ UiDanger (red palette). An *unknown* reason fails **safe to danger** (§11.4.6 — never paint green/neutral on an unrecognized down).

7.3 The connect controller (the only place lifecycle is commanded)

```

final connectControllerProvider =
    AsyncNotifierProvider.autoDispose<ConnectController,
        void>(ConnectController.new);

```

```

final class ConnectController extends
  AutoDisposeAsyncNotifier<void> {
  @override Future<void> build() async {}

  Future<void> connect({String transport = 'auto'}) async {
    state = const AsyncLoading();
    state = await AsyncValue.guard(() async {
      final cfg = await
        ref.read(sessionProvider).resolveTunnelConfig(); //
        overlayIp/routes/dns/token
      await
        ref.read(tunnelPlatformProvider).startTunnel(cfg); //
        shim creates TUN + links core
    });
  }
  Future<void> disconnect() async {
    state = await AsyncValue.guard(() =>
      ref.read(tunnelPlatformProvider).stopTunnel());
  }
  Future<void> toggle(UiConnectionState ui) =>
    (ui is UiConnected) ? disconnect() : connect();
}

```

The ConnectButton is a pure function of uiConnectionStateProvider — no polling, no manual refresh (CI2):

```

final ui = ref.watch(uiConnectionStateProvider);
ConnectButton(
  state: ui,
  onTap: () =>
    ref.read(connectControllerProvider.notifier).toggle(ui),
);

```

7.4 The app-shell UI state machine (diagram)

stateDiagram-v2

```

[*] --> UiDisconnected : app start / no stream value
UiDisconnected --> UiConnecting : connect() → startTunnel →
core Connecting/Handshaking
UiConnecting --> UiConnected : core Connected{transport,rtt}
UiConnecting --> UiDanger : core Down{reason≠stopped} OR
platform error
UiConnecting --> UiDisconnected : platform permissionDenied
UiConnected --> UiReconnecting : core Reconnecting (drop)
UiConnected --> UiDisconnected : disconnect() → core
Down{stopped}
UiConnected --> UiDanger : platform revoked (OS/admin kill –
overrides core)
UiReconnecting --> UiConnected : core Connected (re-dial ok)
UiReconnecting --> UiDanger : core Down{ladder-exhausted/auth-
failed/...}
UiDanger --> UiConnecting : connect() retry (start() again)

```

```
UiDanger --> UiDisconnected : user dismiss + core
Down{stopped}
```

7.5 Offline / optimistic honesty (the no-lie rule)

Access/Connector keep last-known status + a local *intent* (user wants connected) so a flaky control channel never makes the UI lie [04_CLI §8.4, CI2]. **The core's status stream is always the source of truth for *actual* protection state**; the UI distinguishes intended from actual (UiReconnecting) and never paints green on intent alone. A UiDanger (leak / kill-switch tripped / revoke) **overrides any intent** and paints red immediately (§7.2). The platform revoked/permissionDenied events take precedence over a stale core Connected so an OS-side kill is never masked (O3).

7.6 Console live data (no core_ffi)

The Console watches the control-plane WS/SSE (GET /v1/stream, Volume 3) and folds events (device.online, route.changed, policy.compiled, device.revoked) into Riverpod state, so device lists / topology / audit feed update **live without refresh** [04_CLI §8.3, 04_UI §4.3]. The Console uses helix_api only — there is no tunnelStatusProvider (its helixCoreProvider is NoCore).

8. App lifecycle, permission flow, edge cases

8.1 Cold start (the budget path)

flowchart TB

```
    a["main() → runHelixApp"] --> b["ensureInitialized<br/>(no  
work on UI thread, §9)"]
    b --> c["ProviderContainer overrides<br/>(flavor/capabilities/  
core/platform)"]
    c --> d["runApp(HelixApp)"]
    d --> e["home renders immediately<br/>(deferred non-critical  
providers)"]
    e --> f["uiConnectionStateProvider = UiDisconnected<br/>(no  
stream value yet – §7.2)"]
    f --> g["lazy: tunnelStatusProvider listens<br/>only when  
ConnectScreen mounted"]
```

Cold-start target < 1 s on mid-range mobile [04_CLI §10]: AOT, no dart:ffi symbol resolution at launch (the core is started by the shim only on connect()), deferred non-critical providers. uiConnectionStateProvider is UiDisconnected until the first core event (correct — the tunnel is not up at launch).

8.2 Permission flow state machine

stateDiagram-v2

```
    [*] --> NoPermission
    NoPermission --> Requesting : connect() → shim requests VPN
```

```

permission
  Requesting --> Granted : OS grants (Android prepare()/iOS NE
config saved)
  Requesting --> Denied : OS denies →
PlatformTunnelEvent.permissionDenied (01)
  Granted --> Tunnelling : startTunnel proceeds → core
Connecting
  Denied --> NoPermission : user re-taps connect (re-request)
  Tunnelling --> Revoked : OS/admin kills →
PlatformTunnelEvent.revoked (03)
  Revoked --> NoPermission : tunnel torn down; UI =
UiDanger('revoked')

```

permissionDenied is a **first-class event** (O1), distinct from error — the UI shows a "VPN permission needed" affordance (UiDisconnected + a banner), never a half-open TUN and never a fabricated "connected" (§11.4.6).

8.3 Foreground / background (resource discipline)

```

// helix_domain/lib/src/lifecycle.dart
class HelixLifecycleObserver extends WidgetsBindingObserver {
  HelixLifecycleObserver(this.ref);
  final WidgetRef ref;
  @override void didChangeAppLifecycleState(AppLifecycleState s) {
    switch (s) {
      case AppLifecycleState.resumed:
        ref.invalidate(tunnelStatusProvider); // re-listen →
        core re-pushes latest (06 re-emit)
      case AppLifecycleState.paused:
      case AppLifecycleState.detached:
        // autoDispose + Riverpod-3 pause-when-unlistened drops
        the StreamSink subscription
        // (the OS tunnel keeps running in its own extension/
        service – CI1; only the UI stream pauses)
        break;
      default: break;
    }
  }
}

```

Backgrounding the **UI** does not stop the **tunnel** — the tunnel runs in its own OS extension/service process (CI1), so AppLifecycleState.paused only pauses the Dart status subscription (Riverpod 3.0 pause-when-unlistened + autoDispose, [research-flutter_ffi §6]). On resumed the provider re-listens and the core re-pushes the latest TunnelStatus (the broadcast channel's latest-state semantics, [02_ORCH §4.2/§4.6]).

8.4 Edge-case table

#	Edge case	Dart-side handling	Source
E1	VPN permission denied		O1, §8.2

#	Edge case	Dart-side handling	Source
		permissionDenied event → UiDisconnected + banner; no half-open TUN	
E2	OS/admin revokes tunnel mid-session	revoked event overrides core Connected → UiDanger('revoked')	O3, §7.5
E3	Core emits Down{auth- failed} (device revoked)	_classifyDown → UiDanger('auth-failed') (kill- switch stays closed, [02_ORCH §8.5])	§7.2.1
E4	broadcast lag (RecvError::Lagged) on the FFI stream	frb surfaces only the latest value; the Dart Stream never errors on lag — treat as "re-read latest"	[02_ORCH §4.6]
E5	App process death while tunnelling	OS tunnel survives (separate process, CI1); on relaunch tunnelStatusProvider re- subscribes and renders the live core state, never a stale optimistic "connected"	§7.5, CI2
E6	Web Console invokes a tunnel verb	NoCore/NoTunnelPlatform throw UnsupportedTunnelError; the Console UI never wires a ConnectButton (no Capability.tunnel) so this is unreachable, asserted by a test	§6.2, §11 UT
E7	startTunnel called twice (double-tap)	shim startTunnel is idempotent (O1); the controller's AsyncLoading guard disables the button mid-call	O1, §7.3
E8	Channel error (native crash)	error event → UiDanger('error', detail) with the honest native detail string (§11.4.6 — never "probably fine")	O3, §10

9. Size / memory / startup budgets (app-shell scope)

The brief demands tiny, fast, stable [04_UI §10, 05_YBO]. These are the **app-shell** (Dart/Flutter) budgets; the tunnel-core RSS budget (iOS NE G3 ceiling) belongs to the Rust core / shim doc and is the reason the core is Rust not Go [04_CLI §10, 04_ARCH §5.6].

Budget	Target	App-shell means
Mobile install size (Access)	< ~15–20 MB per ABI	--split-per-abi, tree-shake icons/fonts (§5.3), no bundled video, lean staticlib (LTO+strip)
Cold start		

Budget	Target	App-shell means
	< 1 s mid-range mobile	AOT; no core start at launch (§8.1); deferred non-critical providers; const widgets
Idle app memory	single-digit → low-tens MB	Flutter AOT (never Electron, CI6); autoDispose status stream (§3.5)
Frame budget	60/120 fps, no jank	const widgets, RepaintBoundary on the animated ConnectButton, motion only on state change
Web (Console)	fast first paint	Wasm/CanvasKit, route-level deferred loading, no core_ffi_weight (§6)
Console heavy tables/topology	never bloat Access/Connector	deferred-loaded (deferred as), different flavor anyway [04_UI §8]

Non-negotiable (CI6): never Electron, never a webview-wrapped app for native targets — Flutter AOT + Rust core is the only way the size/speed targets are reachable on eight platforms at once [04_CLI §10, 04_ARCH §5.6].

10. Error handling taxonomy (Dart side)

The Dart layer faces three error sources; each maps to a deterministic UI outcome (never a guess, §11.4.6):

Source	Dart type	Trigger	UI outcome
frb FFI call	AnyhowException (frb)	Rust start/stop/exits/setShields returned Err	AsyncError in the controller → snackbar with the honest message; state stays UiDisconnected/last error event semantics
MethodChannel	PlatformException(code, message, details)	native shim threw / channel not implemented	→ UiDanger('error', message) (E8)
EventChannel	PlatformTunnelEvent(error revoked permissionDenied)	native lifecycle	§7.2 projection (E1/E2/E8)
capability misuse	UnsupportedTunnelError	a NoCore/NoTunnelPlatform verb invoked (web/Console)	unreachable by construction; asserted by test (E6)

```
// helix_core_ffi/lib/src/errors.dart
final class UnsupportedTunnelError implements Exception {
  const UnsupportedTunnelError();
  @override String toString() =>
    'Tunnel capability not present in this flavor (CI3).';
}
```

Rules. (1) Every error surface carries an **honest** message — a `PlatformException` detail or a `frb` reason, never "something went wrong, probably transient" (§11.4.6). (2) A failed `connect()` leaves the kill-switch posture to the **core** (the Dart layer never assumes the firewall opened) — the UI reflects only the core's next `TunnelStatus` (CI2). (3) No error path paints green; an unrecognized `Down`.reason fails safe to `UiDanger` (§7.2.1).

11. Test points — tied to §11.4.169

Every app-shell workable item declares its required test types from the §11.4.169 closed set; the **ONLY** permitted absence of a warranted type is an honest §11.4.3 SKIP-with-reason, never a silent gap. Four-layer enforcement per §11.4.4(b) applies to every closure. Every PASS ships captured evidence per §11.4.5/§11.4.69/§11.4.107; the native $\rightleftharpoons$ Dart channel is the only untyped seam and is covered on real devices (AS-I8/O5).

Code (§11.4.169)	Type	Concrete app-shell test point	Evidence
UT	unit	(a) FFI contract test — Dart <code>TunnelStatus/Shields/ExitOption</code> mirrors match the Rust <code>helix-ffi</code> types byte-for-byte (5-variant enum, AS-I7), a drift FAILs gen-check; (b) <code>projectUiState</code> truth-table over all 5 core variants $\times$ platform events; (c) <code>_classifyDown</code> over the [02_ORCH §4.4] vocabulary; (d) <code>NoCore</code> throws <code>UnsupportedTunnelError</code> (E6)	golden fixture diff; truth-table log
IT	integration (real System)	flavor wiring: <code>runHelixApp</code> builds each flavor; <code>helixCoreProvider=RealHelixCore</code> for Access/Connector, <code>NoCore</code> for Console; the <code>MethodChannel</code> marshals <code>TunnelConfig</code> to a real shim (per-OS)	provider-graph dump; channel round-trip log
E2E	end-to-end	the Phase-0 G5 demo: Linux window connect/disconnect toggle + <code>StatusChip</code> going <code>Connecting</code> $\rightarrow$ <code>Handshaking</code> $\rightarrow$ <code>Connected(masque-h3, 23ms)</code> driven <i>only</i> by the core event stream [04_P0 §9]	window-scoped MP4 (§11.4.159)
FA	full-automation (deterministic §11.4.50)	<code>integration_test</code> drives <code>connect</code> $\rightarrow$ <code>status</code> $\rightarrow$ <code>disconnect</code> N=3 identical runs producing the same ordered <code>UiConnectionState</code> trace	3 $\times$ identical state-trace artifacts
CH	Challenges (challenges submodule)	per-flavor Challenge scores the captured connect-journey evidence (not config); asserts the <code>StatusChip</code> text via OCR	Challenge <code>result.json</code>
HQA			

Code (§11.4.169)	Type	Concrete app-shell test point	Evidence
	HelixQA (helix_qa submodule)	autonomous QA session drives connect / exit-pick / shields-toggle across flavors	HelixQA session evidence
WIDGET(UT)	widget (flutter_test)	each helix_design component renders per UiConnectionState (ConnectButton across every state incl. UiDanger); Semantics label asserted (CI5)	golden frames + semantics dump
GOLDEN(UI)	golden (golden_toolkit)	design system stable across themes/ breakpoints (light+dark, compact/ medium/expanded)	golden PNGs
UX	UX	AdaptiveScaffold BottomNav[?]Rail[?]extended-rail transition on resize; touch targets ≥ 48 dp; ⌘K palette on desktop/web	resize recording
SEC	security (§11.4.10)	the session token / map token never logged; flavorProvider overrides cannot inject a real core into Console; no secret in the marshalled TunnelConfig log	grep-empty proof
SC	stress + chaos (§11.4.85)	EventChannel chaos: duplicate up/ down (O6 idempotent fold), torn re-listen after background (§8.3), revoked mid-Connected (E2), permission-denied mid-connect (E1)	recovery trace
CONC	concurrency	double-tap connect() (E7) — AsyncLoading guard + idempotent shim; concurrent setShields + status fold drop no event	concurrency harness log
RACE	race / deadlock	provider rebuild during a stream emission; autoDispose teardown while a startTunnel future is in flight	Riverpod test log
MEM	memory	app-shell idle RSS (low-tens MB, §9); no per-event heap growth over a 30-min connect/disconnect soak; (the iOS NE core RSS G3 ceiling is the shim doc's MEM, cross-referenced)	/proc/ DevTools RSS sample
BENCH	benchmarking	cold-start < 1 s mid-range mobile (§9); first-frame time; status-fold latency core-emit → pixel	startup/ latency CSV

11.1 Anti-bluff rule for the app shell (§11.4.107/§11.4.158)

A green widget test is **not** proof the user is protected. The **E2E + IT (real device)** layers are the evidence: a window-scoped MP4 (§11.4.159) of the real connect journey on a real device, with the StatusChip reading masque-h3 · 23ms confirmed by OCR/vision validation (§11.4.160). Fake the core/API with the **same**

generated model types so unit tests exercise real contracts [research-flutter_ffi §6 `overrideWithValue`]; but the user-visible claim is earned only by the device recording (§11.4.143 — the real browse→connect journey, not a deep-link shortcut).

12. Phase → task → subtask plan (app shell)

Aligned to the program roadmap [SYN §4, 04_P0/P1/P2/P3]. Each subtask is a workable item (§11.4.93 SQLite SSoT).

Phase 0 — prove the Dart seams (the spike) [04_P0]

- **T0.1 frb FFI boundary (gate G5).** S0.1.1 frb generate → frb_generated.dart; S0.1.2 thin HelixCore wrapper + 5-variant TunnelStatus mirror (AS-I7); S0.1.3 Flutter-Linux window: connect/disconnect toggle + StatusChip going Connecting → Handshaking → Connected(masque-h3, 23ms) driven only by the Rust stream. **G5 pass = that demo recorded** [04_P0 §9].
- **T0.2 melos scaffold.** melos.yaml + 5 packages + gen/gen-check drift gate.
- **T0.3 static-map reconcile in the UI (gate G6).** Drive core.start(mapPath) against /etc/helix/map.json; prove the ConnectButton reflects reconcile outcomes.

Phase 1 — MVP client app shell [04_P1]

- **T1.1 Promote the Dart wrapper** to the Phase-1 field set (exits/setExit/setShields/advertise/attachTun); regenerate + drift-gate.
- **T1.2 helix_design v1** — tokens, HelixTheme, the 7 signature components, light+dark, golden tests.
- **T1.3 Riverpod data layer** — tunnelStatusProvider, projectUiState, ConnectController, helix_api + WS client; offline/intent honesty (§7.5).
- **T1.4 Three flavors** — runHelixApp + capability gating; Access / Connector / Console (Web build via §6 conditional import).
- **T1.5 TunnelPlatform Dart impls** — ChannelTunnelPlatform, WindowsPipeTunnelPlatform, NoTunnelPlatform; per-OS real-device smoke (AS-I8).
- **T1.6 Enrollment UX** — device-generated WG keypair (private key never leaves), OIDC
 - anonymous device-token paths, QR enroll-token minting in Console.
- **T1.7 Permission + lifecycle** — the §8.2 permission state machine + §8.3 foreground/background discipline.

Phase 2 — parity (app shell) [04_P2]

- **T2.1 ExitPicker multi-hop** — chain builder, jurisdiction labels, RTT sort.
- **T2.2 DAITA + PQ shields** — toggles + honest cost notes; Shields fields wired.
- **T2.3 path indicator** — surface the Connected.path field (decision D-UI-3) once the core emits it; StatusChip renders direct/relay.
- **T2.4 Console policy-as-code** — PolicyEditor effect-diff + version history.

Phase 3 — the forked targets (app shell) [04_P3, research-flutter_ffi §4/§5]

- **T3.1 HarmonyOS NEXT** — OpenHarmony SIG fork build, ArkTS VpnExtensionAbility Dart channel impl, DevEco signing (biggest platform risk; honest SKIP where the SIG port lacks a VPN/permission API).
 - **T3.2 Aurora OS** — OMP fork build, Friflex-bridged C++ backend Dart channel impl, RPM signing; enterprise SKU CI (honest SKIP where the embedder lacks an API).
 - **T3.3 Web WASM MASQUE proxy** — optional in-page browser-scoped proxy in Console (not a system tunnel; CI3, §6.2).
-

13. Open decisions surfaced by this document

Per §11.4.6/§11.4.66 — options + recommendation, never silently resolved. These are *app-shell* decisions (program-level D1–D8 live in doc 00; client-stack D-CLIENT-1..5 in [04_CLI §14]).

#	Decision	Option A	Option B	Recommendation
D-UI-1	State management	Riverpod 3.x	Bloc	Riverpod — best fit for the stream-driven status fold + testable provider overrides [research-flutter_ffi §6]; the §7 layering maps onto Bloc if the team prefers explicit event/state classes [04_CLI D-CLIENT-2]. A — keep the wire enum consistent with [02_ORCH §4.1]; synthesize disconnected/danger in projectUiState (§7.2). Revisit only if multiple consumers need the richer wire shape.
D-UI-2	TunnelStatus richness	5-variant core enum + Dart projection (this doc, AS-I7)	extend the <i>core</i> enum to 7 variants (Disconnected/Danger/path)	A (Phase 2) — the P2P/relay indicator [04_CLI §13 T2.3] needs a core-emitted field; UNVERIFIED until [02_ORCH] adds it. Until then <code>UiConnected.path == null</code> and the
D-UI-3	<code>Connected.path</code> (direct vs relay)	add <code>path: String</code> to the core Connected variant (Phase 2)	derive nothing — omit the chip detail	

#	Decision	Option A	Option B	Recommendation
D-UI-4	Windows Dart channel	named-pipe WindowsPipeTunnelPlatform (unprivileged app [?] SYSTEM service)	in-process FFI (requires running Flutter as admin)	chip omits it (no fabrication, §11.4.6). A — never run the UI privileged [04_CLI §5.3]; the service hosts the core + wireguard-nt. UniFFI for native shims, frb for Dart; cbindgen is the fallback where UniFFI is thin (HarmonyOS/Aurora may need raw C) [research-flutter_ffi §2, 04_CLI D-CLIENT-3].
D-UI-5	Native shim binding generator	UniFFI (Swift/Kotlin) + frb (Dart)	cbindgen C header everywhere	

14. Cross-document contracts this document fixes

Contract	Fixed value	Consumed by
Dart FFI wrapper (HelixCore: start/stop/statusStream/exits/setExit/setShields/advertise/attachTun/detachTun)	§3.2	the three apps; the FFI contract test
TunnelStatus Dart mirror = the 5-variant core enum (AS-I7)	§3.3	every reactive widget; projectUiState
UiConnectionState projection (projectUiState / _classifyDown)	§7.2	helix_design palette; the app-shell state machine
TunnelPlatform channel wire (helixvpn/tunnel + helixvpn/tunnel/events, TunnelConfig/PlatformTunnelEvent codec)	§4.2	every native shim (Volume 4 shim doc); device.revoked → revoked (Volume 3)
Flavor + capability model (runHelixApp, Capability.*, NoCore, conditional import)	§5–§6	the three apps; the Console-no-core-ffi / Web-only rule
The "UI is a pure function of the status stream" rule (CI2) + offline-honesty	§7.1/ §7.5	all reactive UI; the protection-honesty guarantee
Web-build gate (if (dart.library.ffi) conditional import ⇒ Console-only Web)	§6.1	the build matrix; CI3

Sources verified

- `final/03-client-core-and-ui.md` [04_CLI] — §0.1 (invariants C11–C16), §2 (melos layout), §3 (FFI surface + TunnelStatus/Shields/ExitOption mirrors), §4 (TunnelPlatform contract O1–O5), §5 (per-platform shim matrix), §6 (flavors + capability gating), §7 (design system + palette), §8 (Riverpod layer + offline honesty), §10 (budgets), §12 (testing), §13 (phase plan), §14 (D-CLIENT-1..5).
- `final/v02-data-plane/orchestrator-and-state.md` [02_ORCH] — §4.1 (the **canonical 5-variant TunnelStatus**), §4.2/§4.6 (broadcast latest-state + receiver contract), §4.4 (Down.reason closed-ish vocabulary), §8.5 (auth-failed closed-and-terminal).
- `final/v02-data-plane/transport-trait.md` [02_TX] — §2 (kind() strings), §4.8 (per-carrier effective_mtu: 1420 plain / 1280 masque), §9 (the §11.4.169 test-type codes reused in §11), I4 (one core, three consumers).
- `04_VPN_CLD/HelixVPN-helix-ui-Flutter.md` [04_UI] — §1 (melos), §2 (flavors + tree-shake), §3 (design system), §4 (Riverpod), §6 (shim matrix + HarmonyOS/Aurora), §7 (responsive), §9 (a11y/i18n/l10n), §10 (budgets), §11 (build/codegen drift).
- `04_VPN_CLD/HelixVPN-Architecture-Refined.md` [04_ARCH] — §5.2 (why Flutter / forks), §5.4 (connector headless), §5.6/§5.7 (size/memory, Web = no TUN), §4.2 (codegen no-drift).
- `04_VPN_CLD/HelixVPN-Phase0-Spike.md` [04_P0] — §9 (G5 frb demo + Rust/Dart skeleton), §10 (static map shape, G6).
- `v09-research/research-flutter_ffi.md` [research-flutter_ffi] — §1 (frb 2.12.0, StreamSink→Stream, version pin), §2 (UniFFI for native shims only; uniffi-dart not production-ready), §3 (staticlib/cdylib packaging), §4 (Aurora omprussia fork), §5 (HarmonyOS OpenHarmony SIG fork), §6 (Riverpod 3.x StreamProvider.autoDispose, pause-when-unlistened, overrideWithValue), §7 (synthesis).
- `v09-research/_SYNTHESIS.md` [SYN] — §4 (phases), §5 (client architecture), §9 (constitution bindings). [05_YB0] operator mandate — eight-platform requirement; tiny/fast/stable bar.

Constitution: §11.4.44 (revision header), §11.4.6 (no-guessing / UNVERIFIED tags, fail-safe Down.reason classification), §11.4.66 (decisions = options + recommendation), §11.4.169 (comprehensive test-type coverage), §11.4.5/§11.4.69/§11.4.107/§11.4.158/§11.4.159/§11.4.160 (captured / window-scoped / vision-verified evidence), §11.4.143 (real connect-journey, not a deep-link shortcut), §11.4.28/§11.4.74 (decoupled reusable packages), §11.4.10 (token-handling / no-leak), §11.4.156 (no active CI — local gen-check sweep).

*End of nano-detail specification — helix-ui (Flutter app shell) (Volume 4, Clients). Pairs with [04_CLI §1/§2/§6/§8] (the overview this deepens), [02_ORCH §4] (the canonical TunnelStatus this mirrors), and the Volume 4 per-platform shim doc (the native bodies this document's channel marshals to). Surfaced decisions: **D-UI-1** state management, **D-UI-2** TunnelStatus richness, **D-UI-3** Connected.path, **D-UI-4** Windows pipe, **D-UI-5** binding generator — all presented, none silently resolved (§11.4.66).*